

Portfolio performance — technical appendix

How the report was cleaned, modelled, calculated, and laid out. Four queries are loaded. Instructions, Returns, and Sheet1 are not.

1. Power Query

Cleaning is done in Power Query. The Excel file is not edited. Autodetect relationships is off.

Hierarchy — 6 rows

Promoted headers, trimmed Region, Country, Sector, and Asset ID, and dropped blank asset rows. One row per asset, A through F. This is the only dimension. A Country Map column in DAX renames USA to United States and UK to United Kingdom so the filled map recognises them. Canada, Australia, and Singapore are unchanged.

BS by Region — 72 rows

- Dropped Category = AUM. AUM in the source is Equity + Debt + 3rd Party, so summing it would double-count. Real estate stays. It is gross asset value, used only on the sector mix chart.
- Rounded the three amount columns to 2 decimals to remove floating-point noise.
- Grouped to Asset ID and Category. Asset E is repeated on the sheet and Asset F has a second plan stub. Summing by asset puts year-to-date, forecast, and plan on one row.
- Unpivoted YTD Actual, Forecast, and Operating Plan into Scenario and Amount. Renamed Forecast to Annual Forecast and Operating Plan to Annual Budget, with a sort column so the charts read year-to-date, forecast, budget.

KPI's — 264 rows

- Skipped the 16 junk rows above the real header. Description was a copy of Asset ID and was removed.
- Grouped to Asset ID. Asset E is repeated, the same way as on the balance sheet.
- Unpivoted the value columns into Metric, TimeView (YTD or Annual), and Scenario (Actual, Budget, or Forecast). Stored variance columns were dropped. Variances are measures, so the income statement can show an arrow instead of a stored number.
- All 11 lines remain: NOI, Interest, Net G&A, FF&E Amort, Recoverable Capital Amort, Current Income Tax, Current Capital Tax, MTM Assets, MTM Debt, Other Gain / Loss, Net Income. Six assets × 11 metrics × 4 scenarios = 264 rows.
- Expense lines are already negative. A positive variance means favourable. The icon logic is not reversed for expenses.

Equity Bridge — 117 rows

Typed the columns and kept rows that have both a region and an attribution category. No join. The grain is Region + Scenario (Jan, Jun, Dec 2019) + Attribution Category. The waterfall uses the Bridge column. Annual Impact is the movement since the prior checkpoint and is not on the waterfall.

2. Relationships

Two relationships. Both are one-to-many, single direction, from Hierarchy to the fact.

From	To
Hierarchy[Asset ID]	BS by Region[Asset ID]
Hierarchy[Asset ID]	KPI's[Asset ID]

The balance sheet is not related to KPI's. Equity Bridge is not related to Hierarchy. Region is not a key: North America is Asset A, Asset B, and Asset C. A relationship on Region would be many-to-many and would repeat the full regional bridge on every asset. The waterfall follows the Region slicer through the Bridge Amount measure instead.

Sort by column: Category by Category Sort, Scenario by Scenario Sort, Metric by Metric Sort, Attribution Category by Sort Order, and the bridge Scenario by Scenario Sort.

3. DAX measures

Every measure sits on Hierarchy. Performance return is net income divided by equity. Forecast net income is divided by annual-forecast equity. Budget net income is divided by annual-budget equity (the source column was Operating Plan).

Balance sheet

```

Amount = SUM ( 'BS by Region'[Amount] )

AUM =
CALCULATE ( [Amount], 'BS by Region'[Category] IN { "Equity", "Debt", "3rd Party" } )

GAV =
CALCULATE ( [Amount], 'BS by Region'[Category] = "Real Estate" )

Equity =
CALCULATE ( [Amount], 'BS by Region'[Category] = "Equity" )

Equity Forecast =
CALCULATE ( [Equity], 'BS by Region'[Scenario] = "Annual Forecast" )

Equity OP =
CALCULATE ( [Equity], 'BS by Region'[Scenario] = "Annual Budget" )

```

Income statement

```
KPI Amount = SUM ( 'KPI's'[Amount] )

YTD Actual =
CALCULATE ( [KPI Amount], 'KPI's'[TimeView] = "YTD", 'KPI's'[Scenario] = "Actual" )

YTD Budget =
CALCULATE ( [KPI Amount], 'KPI's'[TimeView] = "YTD", 'KPI's'[Scenario] = "Budget" )

YTD Var = [YTD Actual] - [YTD Budget]

Annual Forecast =
CALCULATE ( [KPI Amount], 'KPI's'[TimeView] = "Annual", 'KPI's'[Scenario] = "Forecast" )

Annual Budget =
CALCULATE ( [KPI Amount], 'KPI's'[TimeView] = "Annual", 'KPI's'[Scenario] = "Budget" )

Annual Var = [Annual Forecast] - [Annual Budget]

NI YTD Actual = CALCULATE ( [YTD Actual], 'KPI's'[Metric] = "Net Income" )
NI YTD Budget = CALCULATE ( [YTD Budget], 'KPI's'[Metric] = "Net Income" )
NI Forecast   = CALCULATE ( [Annual Forecast], 'KPI's'[Metric] = "Net Income" )
NI Budget     = CALCULATE ( [Annual Budget], 'KPI's'[Metric] = "Net Income" )

NOI YTD Actual = CALCULATE ( [YTD Actual], 'KPI's'[Metric] = "NOI" )
MTM Assets YTD = CALCULATE ( [YTD Actual], 'KPI's'[Metric] = "MTM Assets" )
MTM Debt YTD   = CALCULATE ( [YTD Actual], 'KPI's'[Metric] = "MTM Debt" )
```

Performance return

```
Performance Return Forecast = DIVIDE ( [NI Forecast], [Equity Forecast] )
Performance Return Budget   = DIVIDE ( [NI Budget], [Equity OP] )
Performance Return Var pp   = [Performance Return Forecast] - [Performance Return Budget]
Performance Return YTD =
    DIVIDE ( [NI YTD Actual], CALCULATE ( [Equity], 'BS by Region'[Scenario] = "YTD Actual" ) )
```

Equity bridge and drillthrough title

```
Bridge Amount =
CALCULATE (
    SUM ( 'Equity Bridge'[Bridge] ),
    TREATAS ( VALUES ( Hierarchy[Region] ), 'Equity Bridge'[Region] )
)

Asset Title = SELECTEDVALUE ( Hierarchy[Asset ID], "All assets" )

Country Map =
SWITCH (
    Hierarchy[Country],
    "USA", "United States",
    "UK", "United Kingdom",
    Hierarchy[Country]
)
```

4. Visuals

Three pages. Canvas #F2F2F2. Region, Country, and Sector on page 1 also filter page 2. They do not filter page 3.

Page 1 — Portfolio performance

Visual	Fields
Slicers	Hierarchy Region, Country, Sector
Country map	Location = Country Map
Cards	Performance Return Forecast, Performance Return Budget, Performance Return Var pp
Gauge	Value = Performance Return Forecast. Target = Performance Return Budget. Axis 0 to 20%.
Income statement	Rows = Metric. Values = YTD Actual, YTD Budget, YTD Var, Annual Forecast, Annual Budget, Annual Var. Green arrow if the variance is above zero, red if below. All 11 metrics. No asset on the rows.
AUM stacked column	X = Scenario. Y = Amount. Legend = Category. Visual filter keeps Equity, Debt, and 3rd Party only. Real estate is off this chart.
Real estate mix, 100% stacked column	X = Scenario. Y = GAV. Legend = Sector. GAV is real estate only, so this chart has no category filter.
Return by asset	Axis = Asset ID. Value = Performance Return Forecast. This bar is the drillthrough source.
NOI vs mark-to-market	Axis = Asset ID. Values = NOI YTD Actual, MTM Assets YTD, MTM Debt YTD.

Page 2 — Income statement details

Drillthrough field is Hierarchy[Asset ID], with Keep all filters on. Cards show the asset, country, and sector. The matrix is the same income statement as page 1, so every KPI line is on the page. No other chart.

Page 3 — Equity movements

Waterfall. Category = Attribution Category, sorted Opening through Miscellaneous. Y-axis = Bridge Amount. Scenario slicer on the bridge, single select, saved on Dec 2019. Region slicer on Hierarchy. No country slicer and no sector slicer. With December and all regions selected, opening is \$13.00 billion and the total is about \$15.54 billion. North America alone opens at \$10 billion.